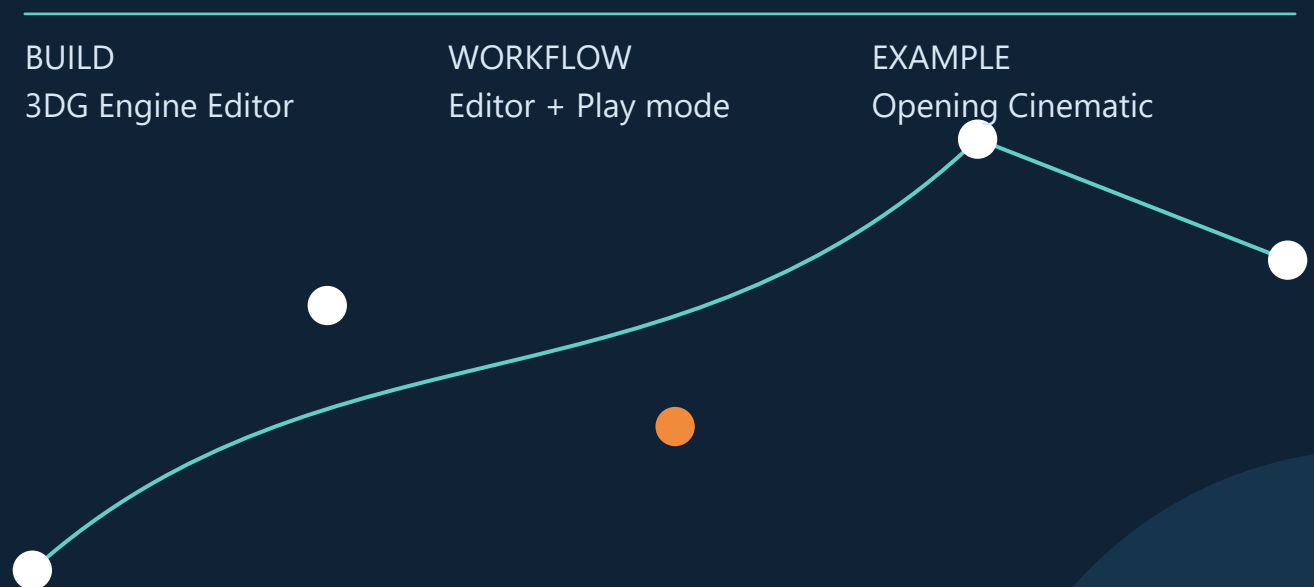


CAMERA MANAGER

Setup and Cinematic Sequence Guide

A hands-on guide to saved cameras, gameplay follow cameras, blends, zones, shake, sequence tracks, triggers, and script control - with a complete Opening Cinematic example.



SECTION 01

Camera system at a glance

The Camera Manager stores cinematic viewpoints in the scene. The Player Controller owns the normal gameplay camera. Camera Zones, sequences, triggers, and scripts decide when another camera behavior takes control.

Where to open it

Use Panels > Camera Manager. The panel is saved with the current scene, so cameras and sequences travel with that scene.

Saved cameras

Reusable position, target, lens, blend, primary, and fixed-Play settings.

Gameplay camera

Player Controller follow camera with collision, shoulder view, and lock-on.

Cinematics

Ordered shots, rails, holds, timeline cues, playback, and script events.

Recommended setup order

- 1** Configure the Player Controller
Establish the everyday first- or third-person camera first.
- 2** Capture named cameras
Frame important views in the viewport, then save them as presets.
- 3** Tune transitions
Choose each camera's blend duration and easing.
- 4** Add zones or a sequence
Use zones for location-based changes and sequences for authored cinematics.
- 5** Connect gameplay
Start sequences from a trigger volume or a script; react to their events.

Control priority: cinematic sequence > active Camera Zone > fixed primary Play camera > Player Controller camera.

The exact runtime choice depends on what is active. Stop or finish a cinematic to return control to the appropriate gameplay camera.

SECTION 02

Create and manage saved cameras

A saved camera is a scene-owned preset. It can be piloted in the editor, used as a fixed Play camera, referenced by Camera Zones, or placed as a shot in a sequence.

Capture a composed view

- 1 Frame the shot
Move the editor viewport to the desired position and aim.
- 2 Capture Current
Creates a camera from the viewport position, forward direction, FOV, and clip planes.
- 3 Rename it
Select the preset and enter a stable descriptive name such as Intro_Wide. Press Enter to commit.
- 4 Verify it
Click Pilot to blend the viewport to the preset. Reframe and use Update From View when needed.

Preset settings

Setting	Purpose	Practical guidance
Position / Target	Location and look-at point.	Keep target away from position; aim at the subject's chest or action focus.
Field of View	Lens angle, 10 to 120 degrees.	35-50 for cinematic shots; 55-75 for responsive gameplay.
Near / Far Plane	Visible depth range.	Avoid an extremely tiny near plane unless close geometry requires it.
Blend Duration	Time used when piloting or entering the preset.	0 for a hard cut; 0.25-1.5 s for most blends.
Blend Easing	Linear, Smooth Step, Ease In, or Ease Out.	Smooth Step is a dependable default.
Use as fixed camera in Play	Allows the primary preset to replace the Player Controller view.	Enable only for fixed-camera gameplay or testing.
Primary	Marks the scene's default saved camera.	Only one preset is primary.

Duplicate is the fastest way to make a nearby angle while preserving lens and blend settings. Renaming a preset updates sequence and zone references.

New Default creates the engine default pose: position (0, 3, 8), target (0, 1, 0), FOV 45, near 0.1, and far 3000.

SECTION 03

Gameplay follow camera

The normal moving-character camera is configured on the Player Controller component. This is separate from saved cinematic cameras and should be tuned before cutscenes.

Core setup

Setting	Default	Effect
First Person	Off	When off, uses an orbiting third-person spring arm.
Eye Height	0.60	Vertical camera origin relative to the player.
Camera Distance	5.00	Desired third-person arm length.
Target Height	1.00	Look-at height above the player origin.
Look Sensitivity	0.10	Mouse or stick look response.
Camera Collision	On	Prevents the camera passing through level geometry.
Probe Radius / Padding	0.20 / 0.08	Collision sweep size and wall clearance.
Return Speed	8.00	How quickly the camera returns after an obstruction clears.

Shoulder camera

- Enable Shoulder Camera for an over-the-shoulder composition.
- Shoulder Offset controls side displacement; the default is 0.65.
- Right Shoulder chooses the starting side. Switch Speed smooths a side change.
- Keep collision enabled. Large offsets make corner clipping more likely.

Lock-on camera

- Enable lock-on only when combat logic can supply a valid target.
- Range limits acquisition distance; View Angle limits the forward search cone.
- Target Height aims above the target origin; Tracking Speed controls camera convergence.
- Defaults are range 18, angle 55 degrees, target height 1, and tracking speed 10.

Tuning rule: Fix collision and return motion before adjusting shoulder offset or lock-on speed. Otherwise camera pops may be mistaken for targeting problems.

SECTION 04

Blends, zones, and camera ownership

Blends make changes readable; zones turn level volumes into camera decisions. Together they support rooms, corridors, fixed-angle encounters, and temporary reveal shots.

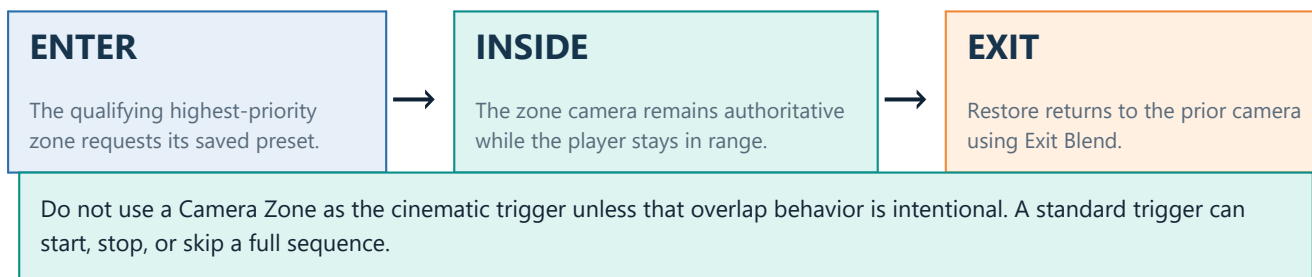
Blend choices

Easing	Motion character	Good use
Linear	Constant interpolation.	Mechanical rails, very short moves, debugging.
Smooth Step	Soft start and soft finish.	General camera changes and most shots.
Ease In	Starts slowly, finishes faster.	Building momentum into a reveal.
Ease Out	Starts quickly, settles gently.	Reaction shots and arrival at a subject.

Set up a Camera Zone

- 1 Create a volume object
Use a primitive sized to cover the intended area.
- 2 Add Collider and Camera Zone
The collider must be configured as a Trigger.
- 3 Choose a saved camera
The zone blends to this preset when entered.
- 4 Choose exit behavior
Enable Restore Camera On Exit and set an Exit Blend.
- 5 Resolve overlaps
Higher Zone Priority wins when zones overlap. Priorities may be negative or positive.

Zone flow



SECTION 05

Camera shake

Camera shake is layered over the active camera, so impacts can be added without replacing the follow camera, zone camera, or cinematic pose.

Preview in Camera Manager

Control	Meaning	Starting point
Duration	Lifetime of the shake.	0.15-0.40 s impact; 0.8-2.0 s rumble.
Frequency	Oscillation rate.	18 Hz impact; 6-12 Hz heavy motion.
Position Amplitude	Local right / up / forward displacement.	Keep under 0.10 for ordinary hits.
Rotation Amplitude	Yaw / pitch displacement in degrees.	0.5-1.5 degrees for readable impact.
FOV Amplitude	Temporary lens pulse.	0-2 degrees; use sparingly.

Click Preview Shake to evaluate it in the editor and Stop Shake to clear active shake. Multiple impulses can overlap and are mixed.

Script examples

```
ShakeCamera(1.0f, 0.35f, 18.0f);
ShakeCameraAdvanced(0.08f, 1.2f, 0.40f, 16.0f, 0.8f);
```

Melee hit

Short duration, medium frequency, small rotation.

Magic blast

Medium duration with a subtle FOV pulse.

Environment

Longer, lower-frequency movement with minimal rotation.

Comfort check: Test shake at the final gameplay FOV and frame rate. Reduce rotation first if the effect feels uncomfortable or obscures aiming.

SECTION 06

How cinematic sequences work

A sequence is an ordered list of saved-camera shots plus timed cues. Each shot first travels to its camera, then holds. The sum of all travel and hold times is the sequence duration.

Shot anatomy

Field	What it controls
Shot Camera	The saved camera reached by this shot.
Travel Time	Seconds spent moving from the previous pose to this camera.
Hold Time	Seconds kept at the camera after travel completes.
Shot Easing	Interpolation profile for the travel portion.
Rail Path	Linear Rail or Catmull-Rom Spline.
Shot Event	Named event fired when the camera reaches the shot.

Rail paths

Linear Rail

Moves directly between poses. Predictable and suitable for cuts, short dollies, and precise blocking.

Catmull-Rom Spline

Uses neighboring camera positions to form a curved rail. Best with several well-spaced shots.

Timeline cues

Track	Required fields	Runtime effect
Event	Cue Time, Event Name	Makes a named event available to gameplay scripts.
Audio	Cue Time, Audio Asset, Volume	Plays the chosen audio asset at the authored time.
Animation	Cue Time, Animation Target, Clip	Plays a clip on the selected animated object in Play mode.

Shot Event vs Event cue: a shot event is anchored to arrival at a camera. An Event timeline cue is anchored to an absolute sequence time. Use the one that expresses the intended dependency.

SECTION 07

Author a sequence in the editor

Create the saved cameras first, then arrange them into shots. Preview repeatedly while editing travel, holds, paths, and cues.

- 1** Create the sequence
In Cinematic Sequences, click New Sequence. Rename it and decide whether it should loop.
- 2** Add shots
Click Add Shot, choose a saved camera, and set Travel Time, Hold Time, Shot Easing, Rail Path, and optional Shot Event.
- 3** Order the shots
Use Move Up and Move Down. Use Remove Shot when a beat is no longer needed.
- 4** Add timeline cues
Use Add Event, Add Audio, or Add Animation. Set each Cue Time inside the authored duration.
- 5** Preview
Enable Show Rails In Viewport, click Play Sequence, then Pause, Resume, Stop, or scrub the Timeline slider.
- 6** Validate Play mode
Test animation targets, audio paths, input locking, skipping, completion, and the return to gameplay camera control.

Sequence timing model

TRAVEL 1 1.00 s	HOLD 1 0.50 s	TRAVEL 2 1.50 s	HOLD 2 0.75 s	TRAVEL 3 1.25 s	HOLD 3 1.00 s
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Total duration = 6.00 seconds. Cue times are measured from 0.00 at sequence start.

Curved-rail advice: use at least four useful camera positions, avoid duplicate positions, and preview the rail around tight geometry. Spline motion can bow outside the straight-line corridor.

SECTION 08

Worked example: Opening Cinematic

This example reveals a wizard and enemy, starts an animation and sound, then hands control back to the player. Names are deliberately stable because triggers and scripts reference them.

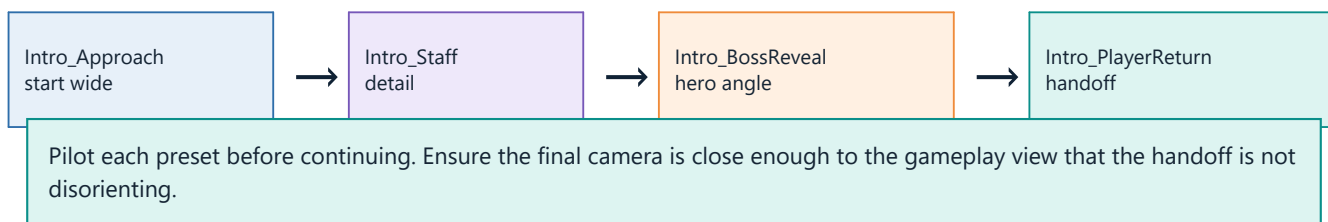
Scene assumptions

- The scene contains a player object named Player and an animated enemy named WizardBoss.
- The boss has an animation clip named Staff_Raise.
- The imported audio asset is Content/Assets/Audio/Magic/Boss_Reveal.wav.
- The Player Controller gameplay camera is already tuned.

Capture these camera presets

Preset name	Composition	FOV	Blend
Intro_Approach	Wide view along the path toward the arena.	50	0.60 s, Smooth Step
Intro_Staff	Medium close-up of the magical staff.	38	0.45 s, Ease Out
Intro_BossReveal	Low three-quarter view of WizardBoss.	42	0.65 s, Smooth Step
Intro_PlayerReturn	View aligned near the normal gameplay angle.	60	0.35 s, Ease Out

Camera placement map



SECTION 09

Example sequence and timeline

Create a sequence named Opening Cinematic. Leave Loop Sequence disabled. Add the four shots in this exact order, then add the timed cues.

Shot list

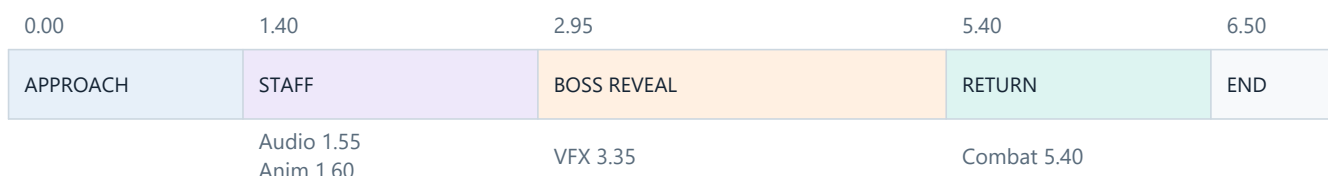
#	Camera	Travel	Hold	Easing	Path	Shot Event
1	Intro_Approach	0.80	0.60	Smooth Step	Linear	ApproachReady
2	Intro_Staff	1.10	0.45	Ease Out	Catmull-Rom	StaffFocused
3	Intro_BossReveal	1.35	1.10	Smooth Step	Catmull-Rom	RevealBoss
4	Intro_PlayerReturn	0.85	0.25	Ease Out	Linear	ReturnControl

Authored duration: 6.50 seconds.

Timeline cues

Time	Track	Configuration	Purpose
1.55 s	Audio	Boss_Reveal.wav, volume 1.00	Begins the magical reveal sound.
1.60 s	Animation	WizardBoss / Staff_Raise	Raises the staff as the detail shot arrives.
3.35 s	Event	EnableBossVFX	Lets a script start staff fire or other magic.
5.40 s	Event	PrepareCombat	Enables combat UI or enemy behavior before handoff.

Timeline



Preview with Show Rails In Viewport. Scrub to 1.55, 3.35, and 5.40 seconds to inspect camera composition; verify cue execution in Play mode.

SECTION 10

Start the sequence from a trigger

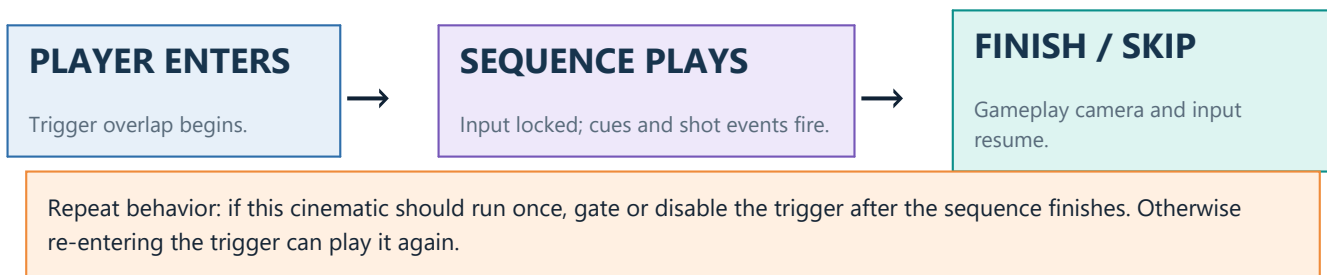
A scene trigger can play, stop, or skip a named sequence on enter or exit. This is the simplest no-code way to begin a cinematic when the player reaches an area.

- 1** Create the trigger volume
Add and scale a primitive at the arena entrance.
- 2** Configure collision
Enable its Collider and set it as a Trigger so it detects overlap without blocking the player.
- 3** Open Cinematic Trigger
In the object's Inspector trigger settings, choose Opening Cinematic.
- 4** Choose Enter Cinematic: Play
Leave Exit Cinematic as None unless the design explicitly stops or skips when leaving.
- 5** Lock Player Input
Enable this so movement cannot break the authored framing.
- 6** Allow Enter To Skip
Enable this to let the player skip with Enter during Play mode.

Trigger configuration

Field	Example value
Cinematic Sequence	Opening Cinematic
Enter Cinematic	Play
Exit Cinematic	None
Lock Player Input	Enabled
Allow Enter To Skip	Enabled

Runtime flow



SECTION 11

Control the sequence from scripts

Scripts can play, stop, or skip sequences, query playback state, and consume named sequence events, completion, and skip notifications.

Minimal scripted sequence controller

```
class OpeningSequenceScript : public engine::Script {
public:
    void OnCreate() override {
        PlayCameraSequence("Opening Cinematic", true, true);
    }

    void OnUpdate(float dt) override {
        if (WasCameraSequenceEvent("Opening Cinematic", "EnableBossVFX")) {
            const auto boss = FindObject("WizardBoss");
            RestartParticles(boss);
            ShakeCamera(0.65f, 0.30f, 16.0f);
        }

        if (WasCameraSequenceFinished("Opening Cinematic")) {
            // Enable combat, UI, or AI here.
        }
    }
};
```

Available camera helpers

Call	Use
PlayCameraSequence(name, lockInput, skippable)	Starts a named scene sequence.
StopCameraSequence()	Stops playback immediately.
SkipCameraSequence()	Skips the active sequence to its end.
IsCameraSequencePlaying(name)	Checks whether a sequence, or any sequence with an empty name, is active.
WasCameraSequenceEvent(sequence, event)	Consumes or checks a named shot/timeline event for gameplay response.
WasCameraSequenceFinished(name)	Detects normal completion.
WasCameraSequenceSkipped(name)	Detects a player or script skip.
ShakeCamera(...) / ShakeCameraAdvanced(...)	Adds an impulse without replacing the active camera.

The second and third PlayCameraSequence arguments are lockInput and skippable. Use explicit values so cinematic behavior is obvious at the call site.

SECTION 12

Validation and troubleshooting

Test camera work in the viewport and in Play mode. Editor preview validates movement and timing; Play mode validates triggers, input, animation, audio, scripts, and handoff.

Preflight checklist

- Every shot references an existing saved camera and every preset has a unique descriptive name.
- The Player Controller camera works before any sequence begins.
- Near and far planes fit the level scale; FOV changes do not feel abrupt.
- Spline rails remain clear of walls, ceilings, and large moving objects.
- Cue times are within the authored duration.
- Audio asset paths resolve and Animation Targets have the named clips.
- Trigger colliders are marked Trigger and cover the intended entry area.
- Input lock, Enter-to-skip, normal completion, and skipped completion are tested.
- The final shot hands off cleanly to the gameplay camera.

Common problems

Symptom	Likely cause	Fix
Primary saved camera is ignored in Play.	Use as fixed camera in Play is off.	Enable it on the primary preset, or keep using Player Controller by design.
Cannot add a sequence shot.	No saved camera exists.	Capture Current or create New Default first.
Camera Zone does nothing.	Missing trigger collider or missing preset.	Configure Collider as Trigger and select a valid saved camera.
Sequence starts but player still moves.	Input lock disabled.	Enable Lock Player Input in trigger or pass true in script.
Audio cue is silent.	Asset path or volume is wrong.	Paste a valid Content path and test the asset independently.
Animation cue does nothing.	Target or clip name mismatch.	Choose an animated scene object and enter its exact clip name.
Spline takes an unexpected arc.	Control cameras are poorly spaced.	Adjust positions or use Linear Rail for that segment.
Scene returns to an odd view.	Final shot differs greatly from gameplay camera.	Add a return camera near the gameplay pose and shorten its hold.

Save the scene after authoring. Camera presets and cinematic sequences are scene data. Test a save/load cycle before treating a cinematic as complete.

SECTION 13

Production checklist and quick reference

Use this page as the final handoff checklist for a gameplay-ready camera setup and a cinematic that can be maintained as the scene changes.

Gameplay camera

- Player Controller follow distance, target height, and look sensitivity tuned.
- Camera collision enabled and tested in narrow spaces.
- Shoulder offset and switching tested on both sides.
- Lock-on acquisition cone, range, and tracking tested against real enemies.

Saved cameras and zones

- Stable names, sensible lens values, and intentional blend settings.
- Primary and fixed-Play flags match the intended ownership model.
- Zones use trigger colliders, priorities, restoration, and exit blends.

Cinematic sequence

- Shot order, travel, hold, easing, and rail path reviewed.
- Rails previewed in the viewport and checked against geometry.
- Events, audio, and animation cues tested in Play mode.
- Trigger or script start path verified; repeat behavior is intentional.
- Skip and finish paths both restore input and gameplay camera control.

Example values at a glance

Item	Value
Sequence	Opening Cinematic, Loop off
Duration	6.50 seconds
Shots	Intro_Approach -> Intro_Staff -> Intro_BossReveal -> Intro_PlayerReturn
Tracks	Audio 1.55, Animation 1.60, EnableBossVFX 3.35, PrepareCombat 5.40
Trigger	Enter Play, lock input, skippable
Key script reaction	EnableBossVFX -> RestartParticles(WizardBoss) + camera shake

Camera work is complete when the transition into the cinematic, the authored motion, the gameplay cues, and the return to player control all feel like one uninterrupted system.

Engine references used for this guide: Editor Camera Manager, Player Controller camera settings, Camera Zones, Camera Shake, Camera Sequence Player, Camera Director, trigger actions, and Script camera helpers.